Knorex: Turns to Google Cloud Machine Learning Engine to enhance real-time bidding in programmatic advertising

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About Knorex

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Established in 2010, Knorex provides a self-serve digital advertising platform called Knorex XPO that provides a one-stop shop for advertisers and media agencies to run programmatic ad campaigns targeting digital channels. The company also provides fully managed services to deliver end-to-end, performance-based advertising campaigns.

Industries: Technology

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digital advertising campaign auctio

Location: Singapore

Cloud Storage

AI Platform (https://cloud.google.com/ai-platform/)

Compute Engine (https://cloud.google.com/compute/)

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Cloud Storage (https://cloud.google.com/storage/)

Google Cloud Results

- Facilitates processing of hundreds of thousands of bid requests in less than 100 milliseconds each
- Delivers cost savings of up to 30% compared to previous platforms
- Reduces instance setup time from 2 days to 3 hours

Traine deploy machi learning models in 1 hour vs. a few days

Programmatic advertising is transforming the way advertisers and media agencies run digital advertising campaigns these days. By automating the negotiation and purchase of digital advertising, programmatic advertising is gradually replacing traditional practices for manual quotations and time-consuming requests for proposal.

Singapore-headquartered, performance-driven advertising technologies and solutions provider Knorex (https://www.knorex.com/) is an important player in this sector in the Asia-Pacific.

Spun off from Singapore's Agency for Science, Technology and Research (A*STAR) in 2009, Knorex has grown to over 100 employees spread across several countries, including Australia, China, India, Malaysia, Singapore, Thailand, Vietnam, and soon the United States.

XPO a "one-stop shop" for programmatic ad campaigns

Knorex's flagship product is a digital advertising platform called XPO that provides a one-stop shop for advertisers and media agencies to execute programmatic advertising campaigns across various digital channels that target audiences with myriad capabilities.

The platform has already delivered campaigns for prestigious clients in Asia-Pacific, including leading hotel groups, airlines, media agencies, financial services companies, and businesses in other industries.

XPO operates by receiving digital advertising bid requests from media publishers worldwide through various advertising exchanges. For each request, the platform determines whether it should proceed to bid, and if so, at what price. XPO does so by reviewing information in the request, any other campaigns Knorex clients are undertaking, and any key performance indicators or restrictions they may have. The platform also obtains a full view of the available audience to help with the determination.

Less than 100 milliseconds

“For every bid request, we have to process and return the bidding decision in less than 100 milliseconds,” Phu Le, Head of R&D, Knorex, explains. “At any given time, hundreds of thousands of bid requests could be coming into the system, totaling billions per day.”

When conceptualizing XPO, Knorex determined scalability and speed were its key priorities. Knorex thus focused heavily on delivering real-time bidding functionality powered by artificial intelligence and machine learning.

Knorex started operating XPO on a dedicated infrastructure in one location. However, to achieve its longer-term growth and service ambitions, the business needed to run XPO on a highly available infrastructure with market-leading machine learning and artificial intelligence capabilities. Knorex must also minimize latency for users distributed across several countries and regions.

Knorex began working with a multinational cloud service, but soon realized the provider could not fully support its growth and expansion plans. “We needed to use multiple cloud-based platforms to help ensure the availability, reach, and latency met our needs for all the geographic locations we served,” Le says. “We also had to control our costs while attaining the scalability to support growth in clients and traffic.”

“We have expanded our use of Google Cloud Platform because we find it to be high performance and considerably more cost-efficient.”

—Phu Le, Head of R&D, Knorex

Google Cloud Platform best fit for the business

Knorex conducted an evaluation and determined that Google Cloud Platform (https://cloud.google.com/) best met its needs. Beginning in 2014, the business decided to run its real-time bidding system and big data processes and pipelines on Google Compute Engine (https://cloud.google.com/compute/) virtual machine instances in the Taiwan Region.

The success of the pilot project prompted Knorex to add virtual machine instances in the United States and Europe and move its databases into Google Cloud SQL (https://cloud.google.com/sql/). Knorex has also used Google Cloud Storage (https://cloud.google.com/storage/) to store raw logs and processed data.

“We have expanded our use of Google Cloud Platform because we find it to be high performance and considerably more cost-efficient,” Le says.

To further support real-time bidding and enable continuous improvement of its development processes, Knorex then moved key applications into container clusters managed by Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine/).

“We undertook a preliminary assessment and elected to go with Google core machine learning service. Google Cloud Machine Learning Engine demonstrates strong support for the TensorFlow open source machine learning framework and fits our needs.”

—Phu Le, Head of R&D, Knorex

Google strong support for cutting-edge machine learning

Knorex also found Google Cloud Machine Learning Engine (https://cloud.google.com/ml-engine/) to be superior to rival cloud-based machine learning products. “We undertook a preliminary assessment and elected to go with Google core machine learning service,” Le says. “Google Cloud Machine Learning Engine demonstrates strong support for the

TensorFlow (https://www.tensorflow.org/) open source machine learning framework and fits our needs."

The business believed the richness of Cloud Machine Learning Engine features could further help to streamline its workflow and enhance the productivity of its artificial intelligence and data science team. Rather than writing ad hoc scripts for machine learning model training, deployment, evaluation, and performance reporting, the team could automate these processes.

A 30% cost reduction

Adopting Google Cloud Platform has enabled Knorex to meet its technical and business requirements, including providing a compelling opportunity for expansion. The company found that the move delivered an immediate 30% reduction in costs and allowed the business to plow the savings into funding growth and development.

Furthermore, the move to Google Kubernetes Engine enabled developers to set up virtual machine instances and associated ecosystems in as few as three hours rather than up to two days as required previously.

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thousands of independent machine learning models and the ability to automatically deploy and update them regularly when new data comes in. We selected Google Cloud Machine Learning Engine to do this for us.”

—Phu Le, Head of R&D, Knorex

Machine learning model training down to one hour

Automation through Cloud Machine Learning Engine has enabled Knorex to cut machine learning model training and deployment processes from a few days to as little as one hour. This has delivered a vast improvement in XPO’s performance and support for clients.

“As we are running performance marketing for clients and optimizing for performance metrics, we need to be able to look at each and every request and be able to estimate the optimal bidding price for an auction,” Le explains. “In order to do that, we need hundreds to thousands of independent machine learning models and the ability to automatically deploy and update them regularly when new data comes in. We selected Google Cloud Machine Learning Engine to do this for us.”

Google Cloud Platform has enabled Knorex to overcome the challenges of optimization, availability, scalability, and latency and fully process every bidding request through XPO within 100 milliseconds.

The business looks to increase its use of Cloud Machine Learning Engine to further streamline the workflows of the team working on real-time bidding. In addition, Knorex is evaluating the use of Google BigQuery (https://cloud.google.com/bigquery/) to expedite processing as data volumes increase, further enhancing the quality of service to its clients.